

Linux Root File System Structure

In Linux, the file system follows a **hierarchical structure** starting from the **root directory** (/). Below is an overview of the **main directories** under the root (/) and their purposes.

1. Root Directory (/)

- The top-level directory from which all other directories branch.
 - It contains essential system files and directories.
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2. Important Directories Under Root (/)

Directory	Description
/bin	Essential binary executables (e.g., ls, cp, mv, cat).
/sbin	System binaries (e.g., fsck, reboot, iptables) used by root/admin .
/boot	Boot-related files (e.g., Linux kernel (vmlinuz) , bootloader (grub)).
/dev	Device files for hardware (/dev/sda, /dev/tty, /dev/null).
/etc	System-wide configuration files (/etc/passwd, /etc/hosts, /etc/fstab).
/home	User home directories (/home/user1, /home/user2).
/lib	Shared libraries needed for basic system binaries.
/lib64	64-bit libraries (on 64-bit systems).
/media	Mount point for removable media (/media/usb, /media/cdrom).
/mnt	Temporary mount point for file systems (/mnt/data).
/opt	Optional software packages (/opt/google/chrome).
/proc	Virtual directory for process information (/proc/1, /proc/cpuinfo).
/root	Root user's home directory (not the same as /).
/run	Runtime variable data (like process IDs: /run/sshd.pid).
/srv	Data for services (like web servers: /srv/www).
/sys	Virtual filesystem for system information (/sys/devices).
/tmp	Temporary files (/tmp/session.log), cleared on reboot.
/usr	User binaries, libraries, documentation (/usr/bin, /usr/lib).
/var	Variable data: logs, cache, spool (/var/log, /var/tmp).

3. Important Subdirectories

🔗 */usr (User Utilities and Applications)*

- /usr/bin/ → User binaries (gcc, python, git).
- /usr/sbin/ → System administration binaries.

- `/usr/lib/` → Libraries for `/usr/bin/` and `/usr/sbin/`.
- `/usr/local/` → Locally installed software.

🔗 `/var` (Variable Data)

- `/var/log/` → Log files (syslog, dmesg, auth.log).
- `/var/spool/` → Print jobs, mail queues.
- `/var/cache/` → Cached data.

🔗 `/proc` (Process Information)

- `/proc/cpuinfo` → CPU details.
- `/proc/meminfo` → Memory usage.
- `/proc/[PID]/` → Running process details.

🔗 `/dev` (Device Files)

- `/dev/sda` → Primary hard drive.
- `/dev/null` → Discard output.
- `/dev/tty` → Terminals.